

Web-Assotest: a browser-based tool for genetic association analysis under multiple inheritance models

01 October 2026

Summary

Web-Assotest is a browser-based tool for analysing the association between a genetic variant and a binary outcome (cases vs. controls) from a 3x2 table of observed genotype counts (wild-type / heterozygote / homozygote, by case/control status).

The tool simultaneously fits a hierarchy of five nested logistic-regression models — null, dominant, co-dominant (additive), recessive, and genotypic — and presents the resulting odds ratios, 95% confidence intervals, likelihood-ratio test statistics, and AIC values in a single model-comparison diagram, together with a Hardy-Weinberg equilibrium (HWE) check for cases, controls, and the combined sample (Wigginton, Cutler, and Abecasis 2005). Users can optionally fit the same model hierarchy with Firth’s penalized logistic regression (Firth 1993; Ploner et al. 2023) to obtain stable estimates when a genotype count is zero or the data are otherwise separated.

Web-Assotest is implemented in R as a single `flexdashboard`/Shiny application (R Core Team 2024; Chang et al. 2024; Iannone, Allaire, and Schloerke 2024), runs entirely in the browser once deployed, and requires no local software installation, scripting, or statistical training beyond entering the genotype counts into an editable table (see Figure 1).

Non-R-savvy users who do not want to run the application locally can run the application directly in a browser from this URL <http://ekstroem.com:3838/webassotest/>.

Statement of need

Testing whether a genetic variant is associated with a disease or trait from a simple genotype-by-outcome count table is one of the most common analyses in applied genetic epidemiology, and it is one that many applied researchers (clinicians, wet-lab biologists, or students) need to perform without necessarily writing R or Python code themselves.

The statistical subtlety is that “the” genetic effect is not a single quantity: the same data can be tested under a dominant, recessive, co-dominant, or fully genotypic model, and these choices are not interchangeable. A variant that shows a strong dominant effect can show a null recessive effect, and vice versa.

Existing options for this analysis are either large, general-purpose genetics toolchains aimed at genome-wide data (e.g. PLINK) that are overkill and require command-line fluency for a single 3x2 table, or R packages such as `SNPassoc` that require the user to already be working in R. **Web-Assotest** fills the gap between these: a lightweight, model-comprehensive, point-and-click tool for the single-marker case, so that the choice of genetic model is made visible and testable rather than assumed. The exact HWE test of Wigginton, Cutler, and Abecasis (2005) is bundled so that genotyping-quality checks and the association test itself are available in one place. The tool has been used as teaching material for case-control study design and has been applied in the author’s own applied genetics research (see Research impact statement below).

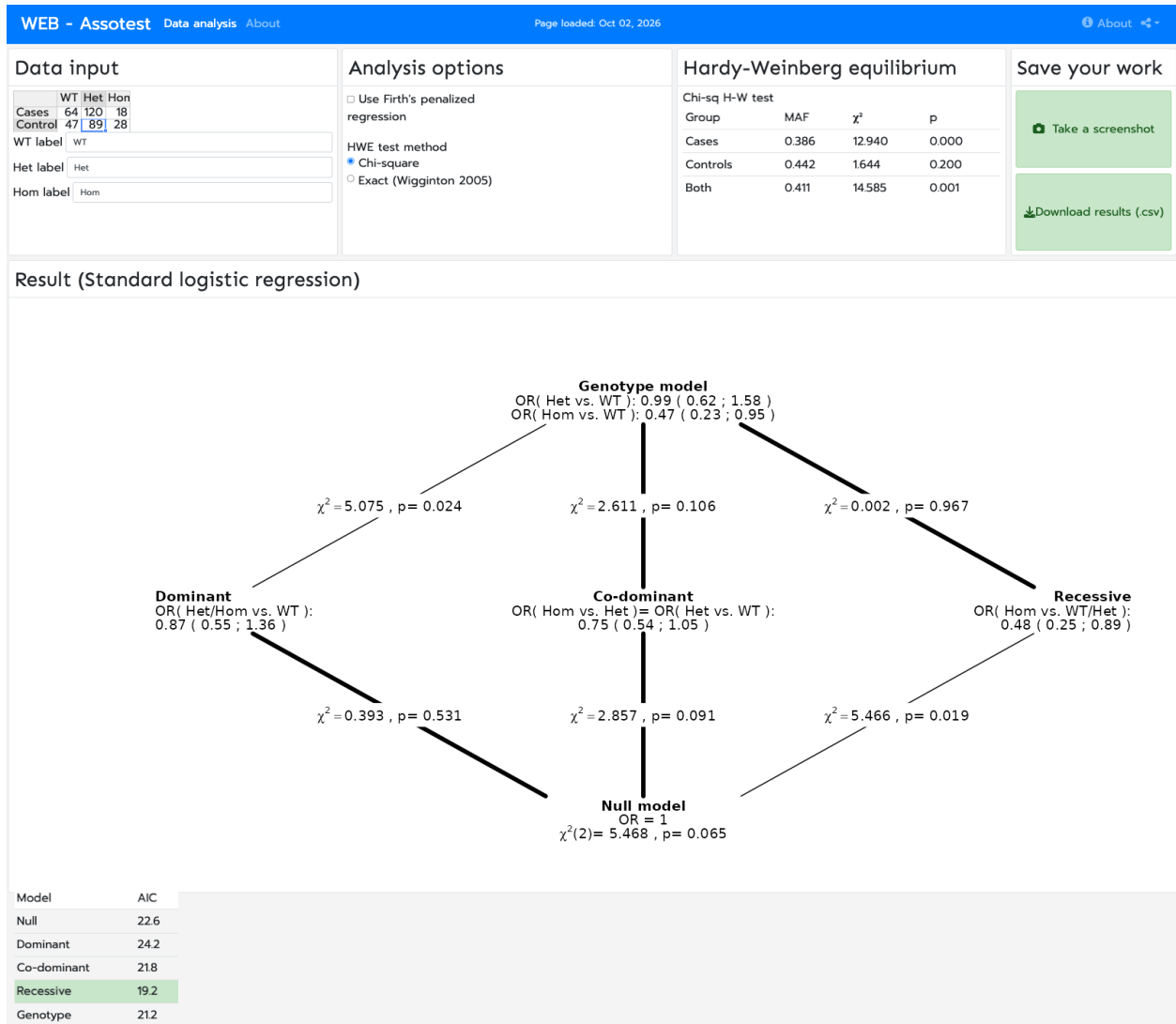


Figure 1: The web-interface for Web-Assotest. User input is entered in the two left-most top panels, and the results are shown in the tables and graph. The result can be saved for documentation in the top right corner.

State of the field

TODO: expand with a short, referenced comparison to alternative tools users might reach for instead of `Web-Assotest`, for example:

- Command-line/whole-genome toolchains (e.g. PLINK) — powerful for genome-wide data but require installation and scripting for a single marker.
- R packages for candidate-gene association (e.g. `SNPassoc`, `genetics`) — flexible but require R programming and are not accessible to non-programming collaborators.
- Standalone HWE calculators (e.g. the Court Lab HWE calculator) — cover HWE testing only, not the genetic-model comparison this tool provides.

`Web-Assotest` differs from all of the above by combining (a) a no-install, browser-based interface, (b) simultaneous fitting and visual comparison of all standard genetic models rather than a single pre-chosen contrast, and (c) an integrated HWE check, in one small single-file application.

Software design

`Web-Assotest` is implemented as a single R Markdown file (`webassotest.Rmd`) built with `flexdashboard` and running as a Shiny application (`runtime: shiny`), so there is no separate `ui.R/server.R` split.

The reactive pipeline reads the user's edits to a 3×2 `rhandsontable` genotype-count table, rounds and validates them, and re-fits all five nested models (`nul`, `gen`, `dom`, `cod`, `rec`) on every change via the function `gtm()`. The application then draws the model-comparison diagram with base R graphics and returns a tidy summary data frame used for both the on-screen Akaike information criterion (AIC) table and CSV export. Likelihood-ratio tests use `anova(..., test = "Chisq")` under standard logistic regression, or a manual deviance difference under Firth's penalized regression (Firth 1993) via the `logistf` package (Ploner et al. 2023); the latter is offered as a toggle so that zero-cell/separated genotype counts do not require the Fisher's-exact-test fallback used in the standard mode. HWE testing (`SNPHWE()`) implements the exact test of Wigginton, Cutler, and Abecasis (2005) directly in R rather than depending on an external genetics package, keeping the application to a single deployable file with a small dependency footprint (`flexdashboard`, `rhandsontable`, `shinyscreenshot`, `logistf`).

Research impact statement

`Web-Assotest` (and its predecessor) has been used in published candidate-gene case-control studies spanning more than two decades (2004–2025) and a broad range of research groups and populations, including Danish, Polish, Iranian, Japanese, Iraqi, and Indian cohorts. The applications cover a wide range of phenotypes and publications, some of which are listed below:

- **Type 2 diabetes and its complications**, including diabetic retinopathy and diabetic foot (S. K. Hansen et al. 2004; S. Hansen et al. 2005; Keshavarz et al. 2006; Mrozikiewicz-Rakowska et al. 2017; Gohari-Lasaki et al. 2020).
- **Obesity, adipokines, and ageing** (Roszkowska-Gancarz et al. 2012; Al-Ogaidi, Abdulsattar, and Al-Dulaimi 2019).
- **Cardiovascular disease**, including essential hypertension and acute coronary syndrome (Wrzosek et al. 2015; Gutowska et al. 2025).
- **Psychiatric and behavioural phenotypes**, including completed suicide, opioid dependence, and paraphilic sexual offending (Chojnicka et al. 2012, 2013, 2014; Fudalej et al. 2016; Jakubczyk et al. 2017).
- **Neurodegenerative disease** (late-onset Alzheimer's disease) (Fatemi Langroudi, Zeinaly, and Ajamian 2023).
- **Autoimmune disease** (Graves' disease) (Kurylowicz et al. 2007).
- **Ophthalmology** (Fuchs endothelial corneal dystrophy) (Ołdak et al. 2015).
- **Gastrointestinal disease** (colonic diverticulosis) (Nehring et al. 2023).

- **Reproductive health** (recurrent miscarriage) (Sudhir et al. 2016).

This breadth illustrates that the tool meets a recurring need in applied genetic epidemiology across clinical specialties: a quick, transparent comparison of genetic inheritance models for a single marker, performed by researchers who are not necessarily statistical programmers.

AI usage disclosure

AI has been used to correct this manuscript for spelling and Claude Code was used to assemble the initial list of papers citing Web-Assotest. The list has subsequently been manually curated. The core part of the `webassotest.Rmd` application predates AI tools and has not been. Claude Code has been used to update and improve the flexdashboard layout and has been used to optimize the load speed of the reactive expressions in the application.

Acknowledgements

I am grateful to Bendix Carstensen who helped form an initial version of the assotest programme. I am also grateful to the users who have provided encouragement throughout the year to make sure that the application was kept available as an online application and continually developed.

References

- Al-Ogaidi, Shahla O, Sura A Abdulsattar, and Hameed MJ Al-Dulaimi. 2019. “FTO Rs17817449 Gene Polymorphism as a Predictor for Maternal Obesity in Iraqi Pregnant Women.” *Indian Journal of Public Health Research & Development* 10 (4): 678.
- Chang, Winston, Joe Cheng, JJ Allaire, Carson Sievert, Barret Schloerke, Yihui Xie, Jeff Allen, Jonathan McPherson, Alan Dipert, and Barbara Borges. 2024. *Shiny: Web Application Framework for r*.
- Chojnicka, Izabela, Sylwia Fudalej, Anna Walczak, Krystyna Wasilewska, Marcin Fudalej, Piotr Stawiński, Katarzyna Strawa, et al. 2014. “Inverse Association Between Obesity Predisposing FTO Genotype and Completed Suicide.” *PLoS One* 9 (9): e108900.
- Chojnicka, Izabela, Krzysztof Gajos, Katarzyna Strawa, Grażyna Broda, Sylwia Fudalej, Marcin Fudalej, Piotr Stawiński, et al. 2013. “Possible Association Between Suicide Committed Under Influence of Ethanol and a Variant in the AUTS2 Gene.” *PLoS One* 8 (2): e57199.
- Chojnicka, Izabela, Katarzyna Strawa, Sylwia Fudalej, Marcin Fudalej, Aleksandra Pawlak, Grażyna Kostrzewa, Marcin Wojnar, Paweł Krajewski, and Rafał Płoski. 2012. “Analysis of Four Genes Involved in the Neurodevelopment Shows Association of Rs4307059 Polymorphism in the Cadherin 9/10 Region with Completed Suicide.” *Neuropsychobiology* 66 (2): 134–40.
- Fatemi Langroudi, SR, Mahsa Zeinaly, and Farzam Ajamian. 2023. “TBX21, the Master Regulator of the Type 1 Immune Response, Overexpresses in the Leukocytes of Peripheral Blood in Patients with Late-Onset Alzheimer’s Disease.” *Immunity & Ageing* 20 (1): 59.
- Firth, David. 1993. “Bias Reduction of Maximum Likelihood Estimates.” *Biometrika* 80 (1): 27–38.
- Fudalej, Sylwia, Andrzej Jakubczyk, Maciej Kopera, Jerzy Piwoński, Wojciech Bielecki, Wojciech Drygas, Krystyna Wasilewska, et al. 2016. “DISC1 as a Possible Genetic Contribution to Opioid Dependence in a Polish Sample.” *Journal of Studies on Alcohol and Drugs* 77 (2): 220–26.
- Gohari-Lasaki, Sahar, Alireza Sharafshah, Saima Abbaspour, and Parvaneh Keshavarz. 2020. “Single Locus and Haplotype Association of ENPP1 Gene Variants with the Development of Retinopathy Among Type 2 Diabetic Patients.” *International Ophthalmology* 40 (3): 639–47.
- Gutowska, Klaudia, Michał Ambroziak, Jakub Podraza, Monika Puzianowska-Kuźnicka, Krzysztof Czajkowski, Andrzej Budaj, and Alina Kuryłowicz. 2025. “Advanced Glycation End-Product Receptor Gene (RAGE) Polymorphisms in Patients with Acute Coronary Syndrome—a Case-Control Study in the Polish Population.” *BMC Medical Genomics* 18 (1): 143.
- Hansen, S. K., A. P. Gjesing, S. K. Rasmussen, C. Glümer, S. A. Urhammer, G. Andersen, C. S. Rose, et al. 2004. “Large-Scale Studies of the Hph i Insulin Gene Variable-Number-of-Tandem-Repeats Polymorphism

- in Relation to Type 2 Diabetes Mellitus and Insulin Release.” *Diabetologia* 47 (6): 1079–87.
- Hansen, SK, CS Rose, C Glümer, T Drivsholm, K Borch-Johnsen, T Jørgensen, O Pedersen, and T Hansen. 2005. “Variation Near the Hepatocyte Nuclear Factor (HNF)-4 α Gene Associates with Type 2 Diabetes in the Danish Population.” *Diabetologia* 48 (3): 452–58.
- Iannone, Richard, JJ Allaire, and Barret Schloerke. 2024. *Flexdashboard: R Markdown Format for Flexible Dashboards*.
- Jakubczyk, Andrzej, Aleksandra Krasowska, Marcin Bugaj, Maciej Kopera, Anna Klimkiewicz, Agata Łoczewska, Aneta Michalska, et al. 2017. “Paraphilic Sexual Offenders Do Not Differ from Control Subjects with Respect to Dopamine- and Serotonin-Related Genetic Polymorphisms.” *The Journal of Sexual Medicine* 14 (1): 125–33. <https://doi.org/10.1016/j.jsxm.2016.11.309>.
- Keshavarz, Parvaneh, Hiroshi Inoue, Yukiko Sakamoto, Kiyoshi Kunika, Toshihito Tanahashi, Naoto Nakamura, Toshikazu Yoshikawa, Natsuo Yasui, Hiroshi Shiota, and Mitsuo Itakura. 2006. “No Evidence for Association of the ENPP1 (PC-1) K121Q Variant with Risk of Type 2 Diabetes in a Japanese Population.” *Journal of Human Genetics* 51 (6): 559–66.
- Kurylowicz, A, YUJI Hiromatsu, B Jurecka-Lubieniecka, Dorota Kula, Małgorzata Kowalska, Michiko Ichimura, Hirohisa Koga, et al. 2007. “Association of NFKB1- 94ins/Del ATTG Promoter Polymorphism with Susceptibility to and Phenotype of Graves’ Disease.” *Genes & Immunity* 8 (7): 532–38.
- Mrozikiewicz-Rakowska, Beata, Agnieszka Sobczyk-Kopciół, Konrad Szymański, Piotr Nehring, Patryk Szatkowski, Joanna Bartkowiak-Wieczorek, Anna Bogacz, et al. 2017. “Role of the Rs2274907 Allelic Variant of the ITLN1 Gene in Patients with Diabetic Foot.” *Polish Archives of Internal Medicine* 127: 319–27.
- Nehring, Piotr, Grażyna Gromadzka, Miłosz Jastrzębski, and Adam Przybyłkowski. 2023. “Genetic Variants in Matrix Metalloproteinases MMP3 (Rs3025058) and MMP9 (Rs3918242) Associated with Colonic Diverticulosis.” *Medicina* 59 (11): 2031.
- Oldak, Monika, Ewelina Ruszkowska, Monika Udziela, Dominika Oziębło, Ewelina Bińczyk, Aneta Ścieżyńska, Rafał Płoski, and Jacek P Szaflik. 2015. “Fuchs Endothelial Corneal Dystrophy: Strong Association with Rs613872 Not Paralleled by Changes in Corneal Endothelial TCF4 mRNA Level.” *BioMed Research International* 2015 (1): 640234.
- Ploner, Meinhard, Daniela Dunkler, Harry Southworth, and Georg Heinze. 2023. *Logistf: Firth’s Bias-Reduced Logistic Regression*.
- R Core Team. 2024. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing.
- Roszkowska-Gancarz, Małgorzata, Zbigniew Bartoszewicz, Jacek Polosak, Alina Kurylowicz, Marta Jonas, Małgorzata Mossakowska, Edward Franek, and Monika Puzianowska-Kuźnicka. 2012. “Total and High Molecular Weight Adiponectin and Level-Modifying Polymorphisms of ADIPOQ in Centenarians.” *Endokrynologia Polska* 63 (6): 439–46.
- Sudhir, Neha, Badaruddoza, Archana Beri, and Anupam Kaur. 2016. “Association of Tumor Necrosis Factor-Alpha 308G/a Polymorphism with Recurrent Miscarriages in Women.” *Journal of Human Reproductive Sciences* 9 (2): 86.
- Wigginton, Janis E, David J Cutler, and Gonçalo R Abecasis. 2005. “A Note on Exact Tests of Hardy-Weinberg Equilibrium.” *The American Journal of Human Genetics* 76 (5): 887–93.
- Wrzosek, Małgorzata, Milena Sokal, Anna Sawicka, M Włodarczyk, M Głowala, Michał Wrzosek, Magdalena Kosior, M Talalaj, Przemysław Biecek, and G Nowicka. 2015. “Impact of Obesity and Nitric Oxide Synthase Gene G894T Polymorphism on Essential Hypertension.” *J Physiol Pharmacol* 66 (5): 681–89.